

Student Number:								
-----------------	--	--	--	--	--	--	--	--



2 Real Analysis 2023
A/Prof Elena Berdysheva
Dr Francesco Russo
Deferred Test 2
23 October 2023



Instructions:

- This question paper consists of **13 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the corresponding boxes.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 50 points = 100% (52 points available).

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

Problem 1

Let $(x_n)_{n=1}^\infty$ be the sequence defined recursively by

$$\begin{aligned}x_1 &= 6, \\x_{n+1} &= \sqrt{3 + x_n}, \quad n \in \mathbb{N}.\end{aligned}$$

- (a) Prove that $2 \leq x_{n+1} \leq x_n \leq 6, n \in \mathbb{N}$. (5 points)

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

--

(b) Formulate the Monotone Convergence Theorem. (2 points)

--

(c) Explain how the Monotone Convergence Theorem is applied to show that the sequence $(x_n)_{n=1}^\infty$ converges. (2 points)

--

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

(d) Find $\lim_{n \rightarrow \infty} x_n$. (3 points)

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Problem 2

(a) Consider the series $\sum_{n=0}^{\infty} \frac{1+2^n}{3^n}$. Find an expression for its partial sums. Use this expression to show that the series converges, and find its sum. (3 points)

Student Number:								
-----------------	--	--	--	--	--	--	--	--

(b) For each of the following series, determine whether it is convergent or divergent. Give a justification.

(i) $\sum_{n=1}^{\infty} \frac{(2n)^n}{n!}$ (2 points)

--

(ii) $\sum_{n=1}^{\infty} \left(\frac{n^2 + 2n - 2}{3n^2 + 5n} \right)^n$ (2 points)

--

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

(iii) $\sum_{n=1}^{\infty} \frac{(\cos n)^2}{n + n\sqrt{n}}$

(2 points)

(c) For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent. Give a justification.

(i) $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n} + 1}$

(4 points)

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

(ii) $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ (3 points)

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

Problem 3

Read the following theorem and an outline of its proof.

Theorem. Let $K \subseteq \mathbb{R}$. If K is closed and bounded, then K is compact.

Proof. Let $(x_n)_{n=1}^\infty$ be a sequence in K . Since $(x_n)_{n=1}^\infty$ is bounded, it has a convergent subsequence. Since K is closed, the limit of this subsequence belongs to K , and the statement follows.

Answer the following questions:

(a) Give a definition of closed set. (1 points)

--

(b) Give a definition of bounded set. (1 points)

--

(c) Give a definition of compact set. (1 points)

--

Student Number:								
-----------------	--	--	--	--	--	--	--	--

(d) How do we conclude that $(x_n)_{n=1}^{\infty}$ has a convergent subsequence? Name and formulate the corresponding theorem. (2 points)

--

(e) How do we conclude that the statement follows at the end of the proof? (2 points)

--

(f) Give an example of a set which is compact. Justify that this set is compact. (2 points)

--

(g) Give an example of a set which is not compact. Justify that this set is not compact. (2 points)

--

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Problem 4

Consider the set $S = \left\{ \frac{(-1)^n n}{n+1} : n \in \mathbb{N} \right\} \cup \{1\} \cup [2, 3]$.

(a) Determine, without proof, the set S' of all limit points of S . (1 points)

--

(b) Determine, without proof, the closure $\overline{S}$ of S . (1 points)

--

(c) Determine, without proof, the set of all isolated points of S . (1 points)

--

(d) Is S compact? Justify your answer. (2 points)

--

Student Number:								
-----------------	--	--	--	--	--	--	--	--

Problem 5

(a) Let $f : A \rightarrow \mathbb{R}$ be a function, let a be a limit point of A and $L \in \mathbb{R}$. Define what it means that $\lim_{x \rightarrow a} f(x) = L$. (2 points)

--

(b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Suppose that $\lim_{x \rightarrow 2} f(x) = 3$. Prove by definition that

$$\lim_{x \rightarrow 2} \frac{f(x) + 1}{f(x) - 1} = 2.$$

(6 points)

--

Student Number:									
-----------------	--	--	--	--	--	--	--	--	--

--